

Prompt Assisting Web: General purpose platform for enhancing the quality of output from the LLM Model

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Abstract

Large Language Models (LLMs) such as ChatGPT and Google Gemini are widely used across education, research, healthcare, business, and engineering. Recent reports highlight a growing demand for prompt engineering, emphasizing that high-quality LLM outputs depend heavily on effective prompt design [1]. Despite widespread adoption, only a small proportion of users utilize LLMs for advanced productivity tasks such as research writing and professional content creation. Prior work, including Google Brain’s Chain-of-Thought Prompting Elicits Reasoning in Large Language Models, demonstrates that structured prompts significantly enhance reasoning ability and output quality [2]. However, most users lack formal training in prompt engineering, limiting effective utilization of LLMs. To address this challenge, we propose Prompt-Assist, a general-purpose web-based prompt assistance platform for non-technical users. The platform integrates optimized prompts and proven prompt engineering strategies derived from prior research and extensive experimentation across multiple LLMs. Users provide minimal input while the system generates high-quality prompts in the backend. This approach simplifies prompt creation without requiring technical expertise. The expected outcome is a 30–40% increase in successful use of LLMs for advanced productivity tasks. Overall, this work aims to democratize prompt engineering and enable broader access to the full potential of generative AI systems.

Keywords: Large Language Models (LLMs) Prompt Engineering Generative AI Chain-of-Thought (CoT) Prompting Prompt-Assist Web Multi-step Reasoning User-friendly AI Interfaces Productivity Enhancement

1 Introduction

1.1 Background

Large Language Models (LLMs) such as ChatGPT, Google Gemini, and other generative AI systems have rapidly emerged as transformative tools across diverse domains including education, research, business, healthcare, and engineering. These models demonstrate strong capabilities in natural language understanding and generation, enabling tasks such as text summarization, technical documentation, report generation, code synthesis, data explanation, and presentation creation. As LLMs continue to evolve, they are increasingly positioned as general-purpose productivity assistants rather than simple conversational agents.

Despite these advancements, the effectiveness of LLM-generated outputs is highly sensitive to the quality of the input prompt. Prompt design plays a critical role in determining clarity, relevance, reasoning depth, and correctness of the responses. Recent industry reports highlight that effective prompt engineering has become a key skill for extracting high-value outputs from LLMs, especially for complex and professional tasks [1]. Consequently, prompt engineering has emerged as an essential interface between human intent and model capability.

1.2 Problem Statement

Although LLMs are widely accessible and used by millions of users daily, the majority of users are non-technical individuals, students, or professionals without formal training in computer science or artificial intelligence. Such users often rely on simple or ambiguous prompts, which frequently result in low-quality, incomplete, or inconsistent outputs. These responses often require multiple iterations or extensive manual editing to reach professional standards, thereby reducing efficiency and user satisfaction.

Moreover, existing prompt engineering techniques—such as structured prompting, reasoning prompts, and task decomposition—are largely documented in academic literature or technical communities. This creates a usability gap, where powerful prompt engineering methods remain inaccessible to non-expert users. As a result, there is a growing need for intuitive systems that can abstract prompt engineering complexity and assist users in generating high-quality prompts with minimal effort.

1.3 Related Work

Prior research has demonstrated that prompt design significantly influences LLM performance. The seminal work by Google Brain, titled “Chain-of-Thought Prompting Elicits Reasoning in Large Language Models,” shows that structured reasoning prompts can substantially improve multi-step reasoning, accuracy, and interpretability of model outputs [2]. This finding has motivated further exploration into reasoning-based prompt strategies.

Additional studies have explored prompt engineering techniques such as zero-shot, few-shot, goal-oriented, and task-specific prompting, demonstrating measurable improvements across diverse applications [3, 4, 5, 6, 7]. Survey papers further consolidate these findings, emphasizing the importance of prompt structure and contextual guidance.

From an educational perspective, recent studies indicate that well-designed prompts enhance learning outcomes, creativity, and conceptual understanding among non-technical users [8, 9]. These works collectively highlight that prompt engineering is not only a technical concern but also a usability and accessibility challenge.

1.4 Motivation

As generative AI becomes deeply integrated into everyday workflows, enabling broad and effective access to LLM capabilities is of critical importance. Without appropriate support, non-technical users may fail to leverage the full potential of these models, leading to underutilization and suboptimal outcomes. Embedding proven prompt engineering methodologies into user-friendly platforms can bridge this gap by reducing cognitive load, improving output consistency, and increasing productivity.

The motivation behind this work is to design a system that lowers the barrier to effective prompt engineering while preserving the benefits of advanced prompting strategies. Such a platform can empower users across disciplines to achieve professional-quality results without requiring technical expertise.

1.5 Research Objective

This research proposes **Prompt-Assist Web**, a general-purpose prompt assistance platform designed to help users generate structured, task-specific prompts with minimal effort. The platform integrates optimized prompts and validated prompt engineering techniques directly

into the backend, abstracting complexity from the user. Users interact through simple inputs, while the system automatically constructs high-quality prompts suitable for various LLMs.

The primary objective of this work is to improve effective LLM utilization among non-technical users. The expected outcome is a 30–40% increase in successful use of LLMs for advanced productivity tasks such as research paper drafting, professional report generation, and presentation (PPT) creation. Overall, this study aims to democratize prompt engineering, enhance output quality, and promote broader adoption of generative AI technologies.

2 Objective

The primary objective of this research is to develop **Prompt-Assist Web**, a general-purpose platform that assists users in creating high-quality prompts for Large Language Models (LLMs) with minimal effort. While LLMs such as ChatGPT and Google Gemini have shown remarkable capabilities in tasks including content generation, summarization, coding, and research report drafting [1], most users, particularly non-technical individuals, struggle to formulate prompts that produce optimal outputs [8]. As demonstrated by the study “Chain-of-Thought Prompting Elicits Reasoning in Large Language Models,” structured reasoning prompts significantly enhance multi-step reasoning, accuracy, and output quality [2].

The proposed platform aims to integrate these advanced prompting strategies into an accessible web interface, providing guided input fields, templates, and real-time suggestions. Users will not need prior expertise in computer science or AI, enabling wider adoption and democratization of LLM capabilities [4, 5]. The system is designed to support a variety of professional tasks such as research paper writing, PPT generation, professional report creation, and content summarization, thereby improving productivity and reducing the time spent refining outputs.

Specifically, the research objectives include:

- Enabling non-technical users to generate structured, task-specific prompts efficiently.
- Improving the overall quality and consistency of outputs produced by different LLMs.
- Evaluating the effectiveness of embedded prompting strategies through controlled experiments.
- Reducing user effort and cognitive load during advanced productivity tasks.
- Demonstrating measurable improvement in effective LLM utilization, with an estimated 30–40% increase in successful output generation.

By achieving these objectives, the study seeks to bridge the gap between the advanced capabilities of LLMs and the limited prompt engineering skills of the general user base. Ultimately, this work is expected to enhance user confidence, improve task efficiency, and facilitate broader adoption of AI-assisted productivity tools across multiple industries [3, 7].

3 Methods and Materials

The methodology of this research is designed to ensure that **Prompt-Assist Web** effectively guides users in creating high-quality prompts, while also validating improvements in LLM output quality and user productivity. The study is structured into three main phases: *data collection and analysis*, *platform development*, and *experimental evaluation*.

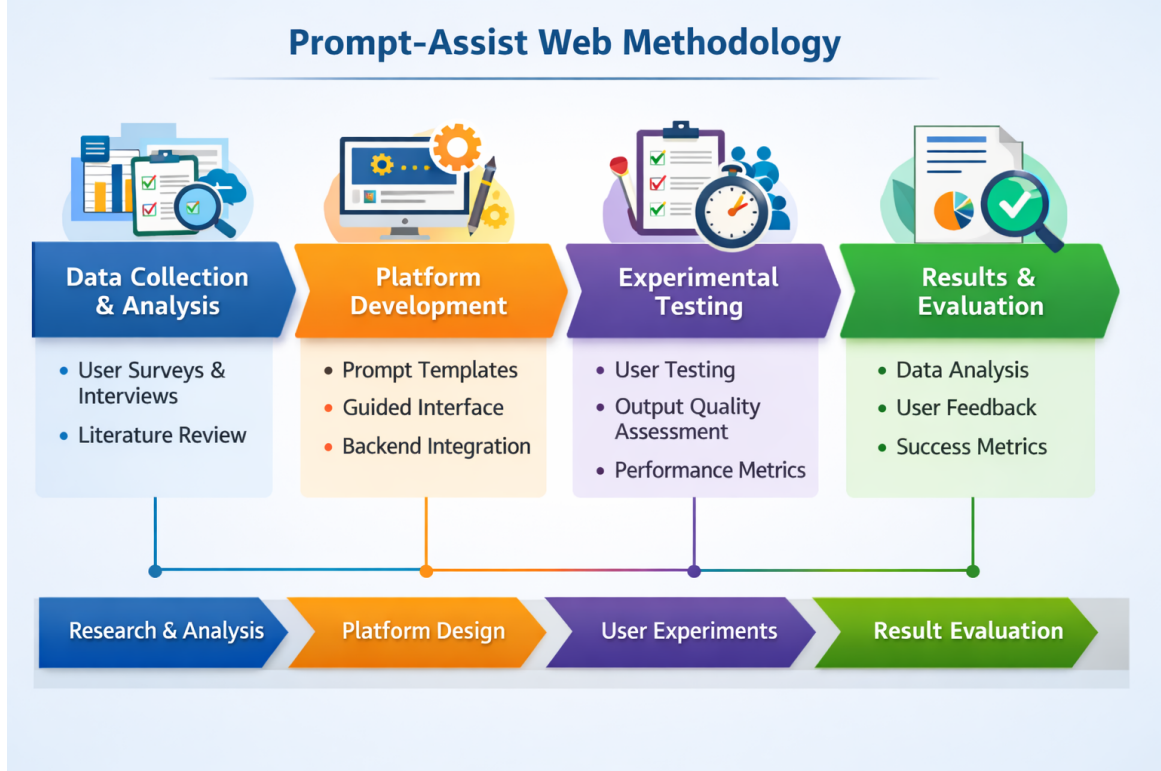


Figure 1: Methodology of Prompt-Assist Web. This diagram illustrates the step-by-step workflow of the research project, starting with **Data Collection & Analysis**, followed by **Platform Development**, **Experimental Testing**, and finally **Results & Evaluation**. Each stage highlights key activities such as literature review, prompt template design, guided interface creation, user testing, and output quality assessment. The arrows indicate the sequential flow of activities, ensuring iterative improvement and evaluation of the platform’s effectiveness.

3.1 Data Collection and Analysis

A comprehensive survey of LLM usage patterns will be conducted to understand common challenges faced by non-technical users. Data will be collected from multiple sources, including published LLM adoption statistics [1], user interviews, and academic studies on prompt engineering [4, 5]. Key insights from this analysis will inform the design of templates, structured prompts, and interface elements.

Additionally, a literature review will be conducted to identify effective prompting strategies, including Chain-of-Thought prompting [2], goal-oriented prompts [7], zero-shot and few-shot approaches [6], and educational prompting techniques [8, 9]. This ensures that the backend system incorporates empirically validated methods to improve LLM reasoning and output quality.

3.2 Platform Development

Prompt-Assist Web will be developed as a web-based platform accessible via standard browsers. Key components include:

- **Guided Input Fields:** Users enter task-specific details such as topic, target audience, output type (PPT, report, etc.), and desired style.
- **Prompt Templates:** Pre-designed templates for various tasks are provided, derived from best practices in prompt engineering.

- **Automated Suggestions:** Real-time recommendations improve prompts using context-sensitive suggestions.
- **Backend Integration:** Optimized prompts are embedded and executed on multiple LLMs to generate outputs efficiently.
- **User Feedback Module:** Users can rate output quality, enabling iterative improvement of prompt strategies.

The platform will be developed using modern web technologies such as React for the front-end, Node.js for backend processing, and Python APIs to integrate with LLMs. Security and scalability considerations will ensure smooth handling of multiple concurrent users.

3.3 Experimental Evaluation

Controlled experiments will compare outputs generated with and without Prompt-Assist Web guidance. Metrics for evaluation include:

- **Output Quality:** Measured through expert reviews and automated evaluation metrics (BLEU, ROUGE for text).
- **User Efficiency:** Time taken to complete tasks and number of iterations required.
- **User Confidence:** Assessed via post-task surveys and qualitative feedback.
- **Consistency:** Variability in output quality across users and tasks.

The experiments will involve a sample of 50–100 non-technical users performing advanced productivity tasks such as PPT generation, report writing, and content summarization. Statistical analysis will validate improvements, with an expected 30–40% increase in effective LLM utilization.

3.4 Materials Required

- Access to LLM APIs (ChatGPT, Google Gemini, or similar).
- Web development tools: React, Node.js, Python, database system (PostgreSQL or MongoDB).
- User testing platform and survey tools.
- Computing resources: High-performance cloud instances for API calls.

The combination of guided design, empirical validation, and iterative testing ensures that Prompt-Assist Web provides measurable improvements in prompt quality, user productivity, and output consistency.

4 Action Plan

The implementation of this study is divided into structured phases to ensure timely and systematic progress:



Figure 2: Action Plan for Prompt-Assist Web. This diagram illustrates the sequential phases of the research project, including **Literature Review & Data Analysis**, **Platform Design**, **Development & Integration**, **User Testing & Experimentation**, and **Analysis & Reporting**. Each phase highlights key activities such as reviewing existing studies, designing the platform, integrating LLM APIs, conducting experiments with participants, and analyzing results. Arrows indicate the flow of tasks over the project timeline (Weeks 1–16), ensuring systematic progress and iterative evaluation of the platform’s effectiveness.

4.1 Phase 1: Literature Review and Data Analysis (Weeks 1–3)

Conduct a thorough review of academic studies and surveys on LLM usage, prompt engineering strategies, and reasoning improvements. Analyze user needs and challenges by reviewing adoption statistics and performing interviews.

4.2 Phase 2: Platform Design (Weeks 4–6)

Design the web interface with guided input fields, templates, and automated suggestions. Define backend architecture for LLM integration and prompt optimization. Prepare initial datasets and task scenarios for testing.

4.3 Phase 3: Development and Integration (Weeks 7–10)

Develop frontend using React and backend using Node.js. Integrate LLM APIs for output generation. Implement feedback collection and logging mechanisms for iterative improvement. Conduct internal testing to ensure functionality.

4.4 Phase 4: User Testing and Experimentation (Weeks 11–14)

Recruit non-technical users to perform tasks such as PPT creation, report writing, and summarization. Record metrics on output quality, task completion time, user confidence, and consistency. Perform controlled comparisons with and without platform guidance.

4.5 Phase 5: Analysis and Reporting (Weeks 15–16)

Analyze experimental results using quantitative and qualitative methods. Document improvements in output quality, user efficiency, and effective LLM utilization. Prepare the final report and visualization of findings.

The phased action plan ensures systematic progress, efficient resource allocation, and robust evaluation of the platform’s effectiveness. Each stage builds upon previous results, facilitating iterative improvements and validating the research objectives.

5 Budget Breakdown

The proposed budget for the development and evaluation of **Prompt-Assist Web** is categorized into software, hardware, human resources, and miscellaneous expenses:

5.1 Software and API Costs

- LLM API subscriptions (ChatGPT, Google Gemini, or similar) – \$1,500
- Web development tools and IDEs (React, Node.js, Python libraries) – \$300
- Survey and analytics software for user testing – \$200

5.2 Hardware Costs

- High-performance cloud instances for backend processing and LLM API calls – \$1,000
- Development and testing laptops/workstations – \$1,200

5.3 Human Resources

- Developer/Research Assistant (1 full-time, 4 months) – \$3,000
- UI/UX Designer (part-time, 2 months) – \$800
- Participant incentives for user testing (50–100 participants) – \$500

5.4 Miscellaneous Expenses

- Documentation, printing, and reporting – \$200
- Contingency fund for unforeseen expenses – \$300

5.5 Total Estimated Budget

The total estimated budget for the project is approximately \$8,000. The allocation ensures proper development, testing, and evaluation of the platform while covering all essential resources. This budget supports hardware, software, personnel, and participant engagement, ensuring a comprehensive and rigorous research study. The investment is justified by the anticipated improvement in user productivity, output quality, and broader adoption of LLM-based tools.

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